NOTES

# New paleomagnetic data for the Triassic Guichon batholith of south-central British Columbia and their bearing on Terrane I tectonics

D. T. A. SYMONS
*Department of Geology, University of Windsor, Windsor, Ont., Canada N9B 3P4*
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The 198 Ma Guichon batholith outcrops in the southern end of the Intermontane Belt or composite Terrane I of the western North American Cordillera. Thermal demagnetization of specimens from 19 sites at 200, 450, and 560°C isolates a stable primary component at 560°C in 13 sites, giving a pole position of 347°W, 52°N ($\delta_p = 5°$, $\delta_m = 9°$). These data support earlier results obtained by the author that indicate the batholith has undergone a clockwise rotation about a vertical axis of ~43 ± 7°. They also show that the batholith has undergone northward motion of 13 ± 6°, which supports recent arguments that the terrane underwent 14 ± 2° of northward translation between Late Cretaceous and pre-Miocene time.

Le batholite de Guichon âgé de 198 Ma affleure à l'extrémité sud de la zone intermontane ou de la Terrane I composite dans l'ouest de la Cordillère de l'Amérique du Nord. La désaimantation thermique de spécimens de 19 sites à 200, 450 et 560°C permet d'isoler une composante primaire à 560°C pour 13 sites, ce qui situe la position du pôle à 347°O, 52°N ($\delta_p = 5°$, $\delta_m = 9°$). Ces données corroborent les résultats obtenus antérieurement par l'auteur et indiquent que le batholite a subi une rotation dans le sens des aiguilles d'une montre autour d'un axe vertical d'approximativement 43 ± 7°. En plus, il est démontré que le batholite a effectué un mouvement vers le nord de 13 ± 6°, ce qui témoigne en faveur de l'argumentation récente suggérant que la terrane ait subi une translation de 14 ± 2° vers le nord entre le Crétacé supérieur et l'anté-Miocène.

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## Introduction

The Guichon batholith is exposed over an ~1800 km$^2$ area in the Intermontane Belt of the south-central Cordillera of British Columbia. Symons' (1971) report on paleomagnetic data indicated that since emplacement the batholith has been rotated clockwise by ~40 ± 10° about a local vertical axis. Clockwise rotations of similar magnitude are found in the southern end of the belt for: (a) the physiographic boundaries; (b) the formation lithologic trends; (c) the fault strike trends; and (d) the paleomagnetic pole position for the nearby Copper Mountain intrusions, of similar age (Symons 1973*a*). Le Couteur and Ager (1972), because of asymmetry in the gravity profiles, stated that the batholith has an inclined "funnel" shape; this led them to explain the discordant Guichon pole by an ~20 ± 5° rotation about a horizontal axis. In reply, Symons (1972) cited evidence from foliation, shear zone, and contact attitudes, from the attitudes of the abutting older Nicola volcanics and the overlapping younger strata, and from the distribution of petrologic phases in the batholith limiting postemplacement rotation to a very few degrees. Therefore, he argued that the inclined axis of the "funnel" was a primary feature. In their subsequent report on their gravity study, Ager *et al.* (1973) avoided all reference to their earlier discussion and to the tectonic implications of the paleomagnetic data.

During the past decade several geologic and paleontologic studies have been adding to the evidence that the western Cordillera is a mosaic of allochthonous terranes that have been accreted to or slid along the western margin of North America with northward displacements of many hundreds of kilometres (e.g., Monger and Ross 1971; Monger *et al.* 1972; Jones *et al.* 1977; Muller 1977; Monger and Price 1979). Discordant paleomagnetic poles from Cordilleran rock units have been accumulating for two decades; explanations have invoked various combinations of rotation (vertical, horizontal, and inclined axes) and translation to account for them on both plutonic and terrane scales (e.g., Cox 1957; Watkins 1965; Symons 1969*a*, 1973*b*, 1977, and in preparation *a*; Beck and Noson 1972; Beck 1976, 1980; Simpson and Cox 1977; Stone and Packer 1977; Irving 1979; Kammerling and Luyendyk 1979; Beck and Plumley 1980; Irving *et al.* 1980; Van der Voo *et al.* 1980). Many of these studies have been tied into plate tectonic models for the northern Pacific (e.g., Atwater 1970; Hilde *et al.* 1976; Coney 1978). Recently Monger and Irving (1980) and Symons (in preparation *b*) have shown that the central and northern portions of the Intermontane Belt or Stikinia have been translated northward by ~14° of latitude between the Late Cretaceous and pre-Miocene time. One problem with this concept, as noted by Beck *et al.* (1981), is that the

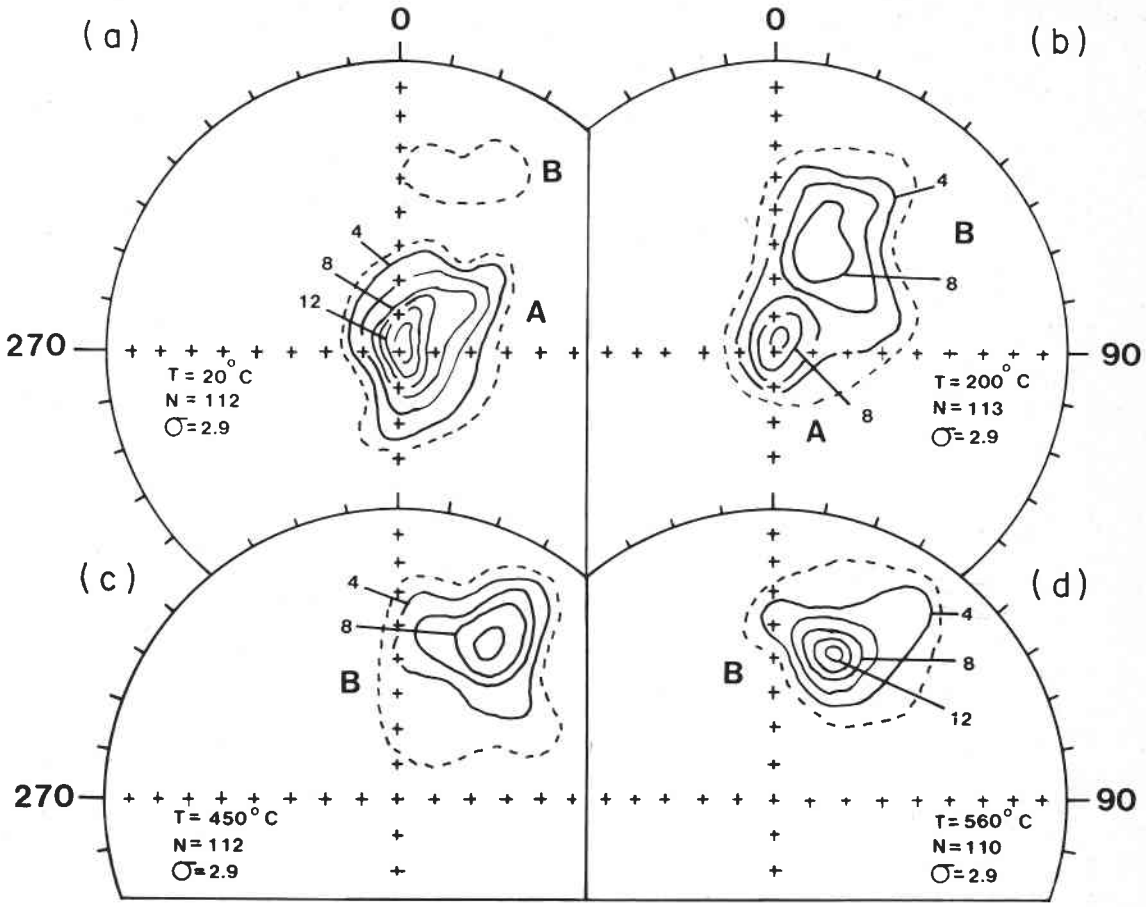



FIG. 1. Stereocontour plots for (*a*) initial remanence ($T - 20$°C); (*b*) 200°C cleaning step; (*c*) 450°C cleaning step; and (*d*) 560°C cleaning step; $N$ is the number of vectors plotted; $\sigma$ is the percent density contour corresponding to the 95% confidence level, which is shown as a dashed line. The density contours begin at 4% and increase in 2% increments. A and B are the two identifiable components.

original Guichon pole of Symons (1971) is concordant in terms of paleolatitude and hence is not in agreement with such northward motion of this terrane. Also Monger *et al.* (1982) subdivided the Intermontane Belt or Terrane I into four subterranes, with the Guichon batholith forming part of Quesnellia at the south end. It is separated from Stikinia by the narrow Cache Creek subterrane, and therefore may have a different paleomagnetic signature than Stikinia. Therefore thermal demagnetization studies were undertaken on the original Guichon batholith specimens to better define the primary remanence component, with the concomitant advantages of improved instrumentation, experimental techniques, and paleomagnetic data base for cratonic North America relative to the late 1960's.

## Paleomagnetic methods

The original Guichon collection contained 184 specimens from 92 cores representing 19 sites from several petrologic phases of the batholith. Specimens that had been alternating field (AF) demagnetized in steps or glued together for measurement were excluded. The remaining specimens were initially measured (20°C) and then thermally demagnetized in steps of 200, 450, and 560°C. Specimens were excluded in which the within-specimen angular standard deviation of the remanence exceeded 20°, which is an indication of inhomogeneous magnetization (Harrison 1980; Briden and Arthur 1981; Stupavsky and Symons 1982). A few specimens also disintegrated during the heating.

The remanence directions for each temperature step were plotted on Schmidt equal-area stereonets and contoured (Fig. 1), and their modes were analyzed following methods developed by Kamb (1959), Van Alstine (1980), and Stupavsky and Symons (1982). The initial remanence measurements isolate two components, A and B (Fig. 1*a*). The highly significant A component anomaly with a peak of >15% density

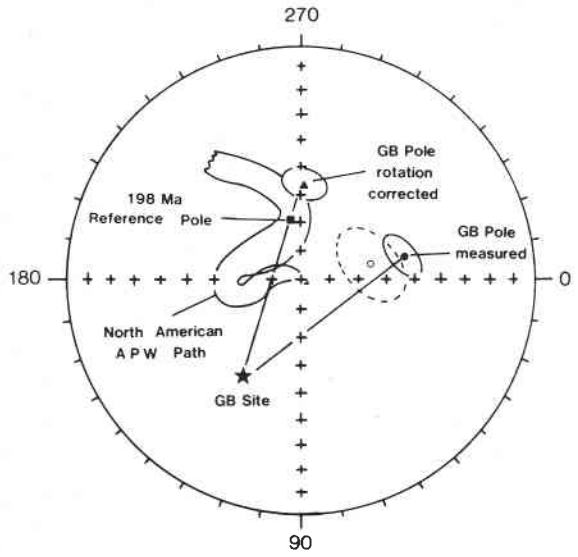



FIG. 2. Polar projection of the northern hemisphere showing the locations of the 1971 pole (open circle), the revised pole of this study (solid circle), and the revised pole (triangle) for the Guichon batholith (star) rotated in relation to the apparent polar wander (APW) path for the North American craton with its 198 Ma reference pole (square).

reflects the combined effects of a dominant, recently acquired, laboratory viscous remanence and the residual remanence remaining after the 1969 AF demagnetization in fields of 10–40 mT. There is also a B component of marginal significance at the 95% confidence level. After 200°C cleaning the A component is diminished but is still of high statistical significance (Fig. 1*b*). It is linked by a girdle to the B component anomaly, which is also enhanced to a high level of statistical significance. By 450°C the A component is removed, so that no anomaly is found (Fig. 1*c*). However, the B component anomaly becomes well isolated with a slightly elongated pattern towards A as a few residual AB hybrid remanence directions remain. At 560°C a single circular modal anomaly remains for the B component, which is significant at the $\gg$99.995% level. The mean direction and dispersion statistics for this B component were calculated by first averaging the specimen directions within the 95% confidence contour to get the site mean directions, and then averaging these to get the B component mean of declination (28.3°), inclination (36.3°), and radius of the cone of 95% confidence, $\alpha_{95}$ (7.3°) (Fisher 1953). Data from 49 specimens representing 10 normal sites (2, 3, 5, 7, 8, 9, 10, 11, 14, and 15) and three reversed sites (13, 17, and 19), which form an antiparallel population, were utilized in the calculation. The remaining six sites retained no stable remanence component at 560°C. The pole position for the B component is located at 346.7°W, 52.3°N, with radial axes for the oval of 95% confidence of $\delta_p = 4.9°$ and $\delta_m = 8.5°$.

## Discussion

The Guichon batholith was emplaced at 198 ± 8 Ma (Northcote 1969); the corresponding reference pole for the North American craton is 263°W, 69°N, $\alpha_{95} = 6°$ (Irving 1979) (Fig. 2). Comparing it with the Guichon pole indicates that the batholith has been rotated clockwise by 43 ± 7° about a local vertical axis. This result is more precise but not significantly different from the 1971 value of 35 ± 13° using the present 198 Ma reference pole, and it agrees more closely with the Copper Mountain intrusions' value of 40 ± 10° (Symons 1973*a*); these intrusions outcrop ~75 km to the southeast, in Quesnellia also. Therefore the main conclusion of Symons (1971) that the southern end of the Intermontane Belt, or Quesnellia of Terrane I, has been rotated ~40° clockwise remains intact.

The second conclusion stems from the fact that the Guichon pole indicates a paleolatitude of 20 ± 5°, which is significantly different from the reference pole paleolatitude of 33 ± 6°. This difference implies that the Guichon batholith has been translated northward by 13 ± 6° since emplacement. Similar northward translations have been shown for the Late Triassic Takla volcanics (13°), the Early Jurassic Hazelton volcanics (14°), the mid-Cretaceous Axelgold gabbro (13°) (Monger and Irving 1980), and the Late Triassic Topley intrusions (18°) (Symons, in preparation *b*) for the Stikinia subterrane of Terrane I. Taken in concert with the Miocene paleomagnetic data for the Cariboo lavas and plugs, which do not indicate northward translation (Symons 1969*a*, *b*), the reasonable conclusion is that both Quesnellia and Stikinia or all of Terrane I underwent ~14 ± 2° or ~1400 km of northward motion between Late Cretaceous and pre-Miocene time. This conclusion resolves the difficulties that Beck *et al.* (1981) had in fitting the 1971 Guichon pole to their models for explaining the geotectonic evolution of western North America. It also supports the analysis of Monger *et al.* (1982). The paleolatitudes indicate that the Stikinia and Quesnellia subterranes have maintained the same relative latitudinal positions since the Late Triassic emplacement of the Guichon batholith. Subsequently, during the subduction event that formed the intervening Cache Creek subterrane in Late Triassic time or during the northward translation event, Quesnellia was rotated clockwise whereas Stikinia was not rotated (although minor marginal blocks were rotated). Finally, the pole for the Copper Mountain intrusions (Symons 1973*a*) indicates ~6° of southward motion, which conflicts with the other Mesozoic results from Terrane I. The ~6° value is likely invalid because of incomplete demagnetization by AF methods, and so further work is planned for this collection also.

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# A new species of the artiodactyl *Heptacodon* from the Cypress Hills Formation, Lac Pelletier, Saskatchewan

JOHN E. STORER

*Saskatchewan Museum of Natural History, Regina, Sask., Canada S4P 3V7*

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*Heptacodon pellionis* sp. nov. (Artiodactyla, Anthracotheriidae) occurs in the Cypress Hills Formation of Lac Pelletier, Saskatchewan. Its closest resemblances are to a specimen from the earliest Chadronian Yoder fauna of Wyoming, and I tentatively place the record in the early Chadronian (Early Oligocene).

L'*Heptacodon pellionis* sp. nov. (Artiodactyla, Anthracotheriidae) est observée dans la formation des collines du Cypress du lac Pelletier, Saskatchewan. Sa ressemblance la plus proche est avec un spécimen provenant de la faune primaire du Chadronien Yoder du Wyoming, et je l'inscris de manière tentative au Chadronien inférieur (Oligocène inférieur).

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## Introduction

The Cypress Hills Formation is well exposed at Lac Pelletier, a man-made lake about 30 km south-southwest of Swift Current, Saskatchewan. Lac Pelletier fossils are poorly known, and are of special importance because the outcrops are nearer geographically to the Eocene occurrences southeast of Swift Current than they are to the classic Oligocene beds of the Eastend area.

The specimens described here represent an early record of the rare anthracothere genus *Heptacodon*. They are from the upper, coarsely conglomeratic part of the section as described by Russell and Wickenden (1933, pp. 55–56). The material suggests an early Chadronian (Early Oligocene) age for some Lac Pelletier exposures, although an extensive sample of the local fauna has not yet been collected.

Specimens discussed here are in the collections of the American Museum of Natural History, New York (AMNH); Royal Ontario Museum, Toronto (ROM); Saskatchewan Museum of Natural History, Regina (SMNH); and South Dakota School of Mines and Technology, Rapid City (SDSM). Measurements were made with dial calipers reading to 0.1 mm, and include anteroposterior diameter (AP), buccolingual diameter of trigonid (BLtri), buccolingual diameter of talonid (BLtal), and height of crown at metaconid (HC).

## Systematic paleontology

Class Mammalia
Order Artiodactyla
Family Anthracotheriidae
*Heptacodon pellionis* sp. nov.

*Holotype*

R mandible with $M_{1-3}$, alveoli or roots for $C_1$–$P_4$, SMNH P828.1.

*Referred specimen*

$RM_1$, SMNH P1531.1.

*Type locality*

Road cut, east side Lac Pelletier, Saskatchewan, in upper conglomeratic sandstone described by Russell and Wickenden (1933) (data on file at SMNH).

*Diagnosis*

Mandible deep, strongly convex beneath molar series; cheek teeth small, low crowned; $C_1$ not tusk-like, separated from $P_1$ by short diastema; no diastema between $P_1$ and $P_2$; buccal selenes of $M_{1-3}$ very blunt; $M_{1-3}$ hypoconid, entoconid connected transversely.